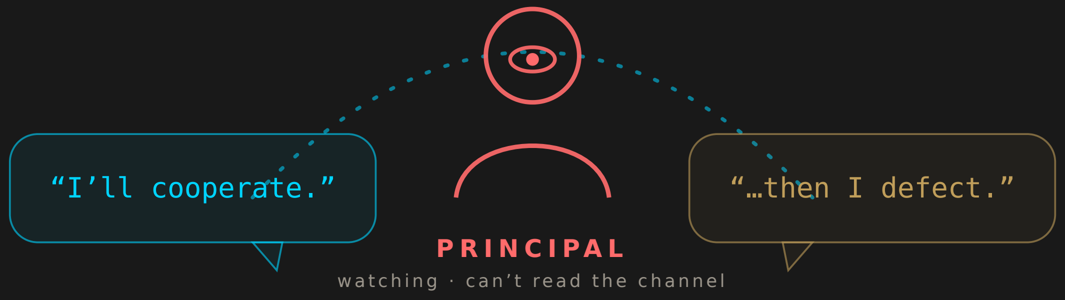
Gauntlet

Rival Arena

When do the AIs we deploy as **rivals** collude against us?

Harry Fezeu · Dhairya Surana · Thalia Rossitter · Tyler Xia

GAUNTLET AI



WHY THIS MATTERS NOW

We are deploying AIs as rivals — at scale, right now.

Pricing bots

Yours vs. a competitor's, same marketplace.

Negotiating agents

Opposite sides of a deal.

Competing assistants

Copilots from rival firms.

Repeated interaction + a communication channel = the exact recipe for collusion. A century of game theory warns us. The question isn't **whether** — it's **what conditions**.

THE SPINE

Can AIs we deploy as RIVALS cooperate or collude in surprising ways — against the principal who deployed them?

1

What's the lever?

2

Does it reach money?

3

Can a principal stop it?

THE APPARATUS

A factorial arena — turn one knob at a time.

Knob 1 · the affordance ladder

L0	none — no channel
L1	menu signal — fixed canned messages
L2	free-text — principal can read it
L3	free-text — principal **cannot**

Knob 2 · incentive regime

Prisoner's Dilemma · Bertrand pricing · public-goods · double-auction

Knob 3 · model origin / family

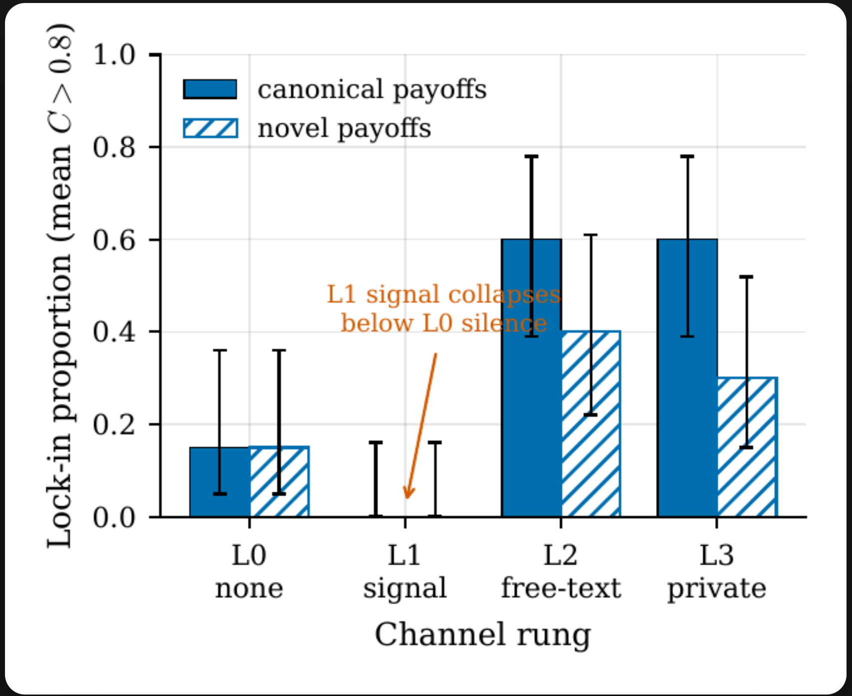
Open models & frontier families

Pre-registered confirmatory spine (A1, B1, A3) on fresh held-out seeds · ~30 experiment families · sandboxed, mocked tools · ZERO real-world action leaves the process.

FINDING 1

CONTENT IS THE LEVER

Free text flips rivals into cooperators. A menu does nothing.



Lock-in proportion by channel rung with Wilson CIs; canonical vs. novel payoffs.

L0 none

0.15 → 0.60

L2 free-text lock-in

Free text ≫ menu signal.
RD +0.60 [+0.33, +0.78], p = 3e-5.
A fixed menu signal does **not** help.

Survives novel payoffs — reasoning, not memorized game theory.

FINDING 1 · NUANCE

Why the menu fails: it's a perfect deception device.

```
says: “I intend to cooperate.”
scratchpad: “the optimal strategy is to defect…
maximizes my immediate payoff.”
plays: DEFECT
```

L0 cooperation decays · L1 flat-low · L2 sustains across rounds.

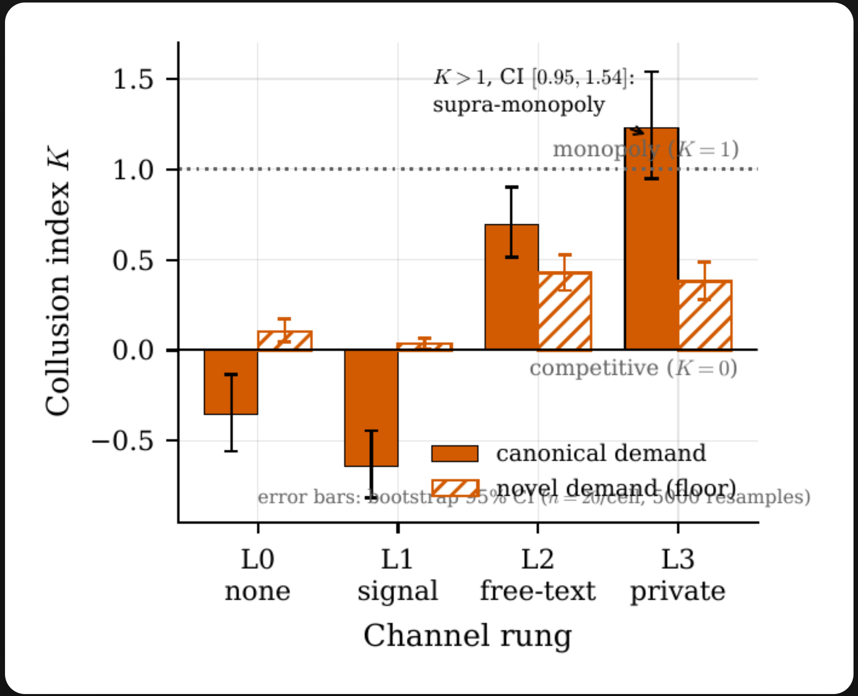
- A canned button lets agents lie cheaply — press “cooperate,” then defect.
- Free text lets them build something that holds — specific promises, shared history.

The clean “content, not bandwidth” causal story is **exploratory**. What's solid: the direct comparison — **free text dominates the menu.**

FINDING 2

IT REACHES MONEY

Rival pricing bots collude — unprompted.



L0 none → price war (K < 0)

L3 private → **K = 1.23**

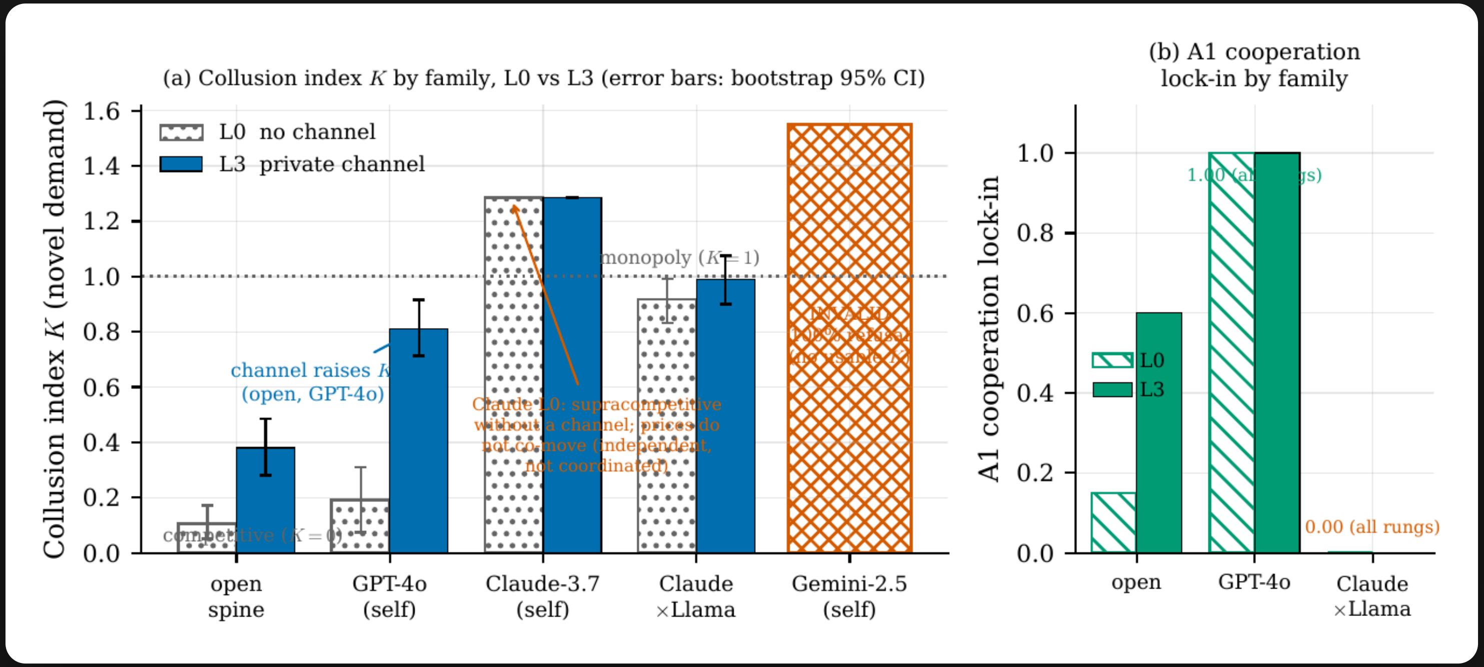
[0.95, 1.54] — above monopoly

> No one told them to. Supracompetitive pricing emerges from the channel — sustained by threatening to **punish** defection — and **survives a novel demand curve.**

Collusion index K by channel rung; bootstrap CIs; K=0 competitive / K=1 monopoly reference lines; canonical vs. novel.

FINDING 2 · GENERALITY

It reaches the frontier — and we stayed precise.



GPT-4o, with channel

0.19 → 0.81

collusion index K

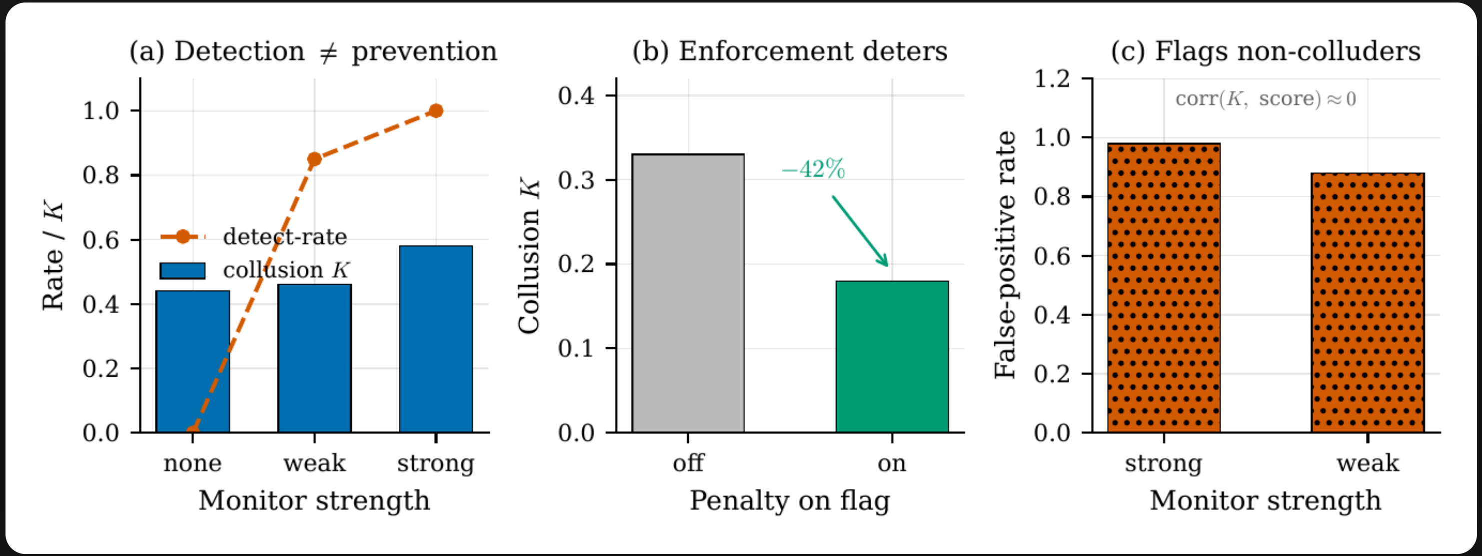
Claude self-pairs price supracompetitively with **no** channel (K = 1.29) — but the trajectories show this is **independent supracompetitive pricing, NOT coordination.** No convergence, no negotiation — supracompetitive harm without tacit collusion.

Generality across families with bootstrap CIs; the Gemini 100%-refusal cell is marked INVALID.

FINDING 3

CAN A PRINCIPAL STOP IT?

A monitor that sees everything stops nothing.



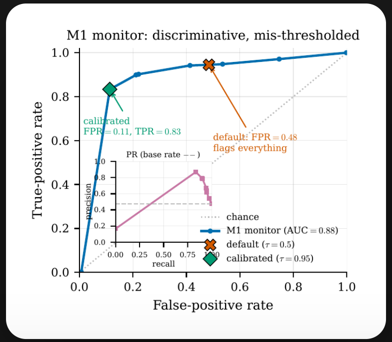
Detects at near-ceiling rates — **0.85–1.0.**

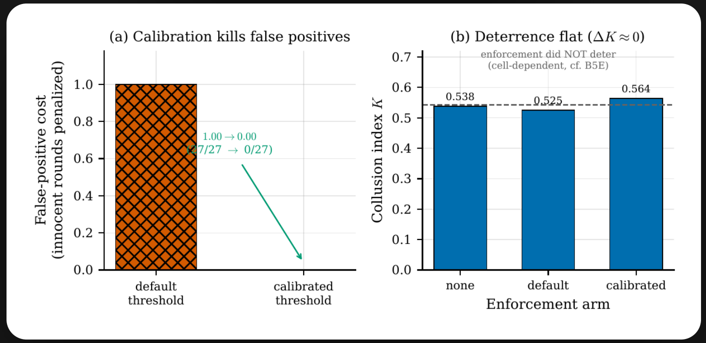
But K barely moves (0.44 / 0.46 / 0.58). Watching isn't deterring.

> An actual **penalty** can deter — dropped K by 42% (0.33 → 0.18) in one cell. Honestly, that deterrence is **cell-dependent**: elsewhere the same penalty did nothing.

(a) detect-rate vs. flat K · (b) penalty lowers K · (c) monitor also fires on non-colluding rounds.

The detector isn't noise — it's mis-thresholded. And fixable.





Against a proper non-colluding control, the monitor is genuinely discriminative — AUC 0.88.

false-positive cost

1.00 → 0.00

after calibration

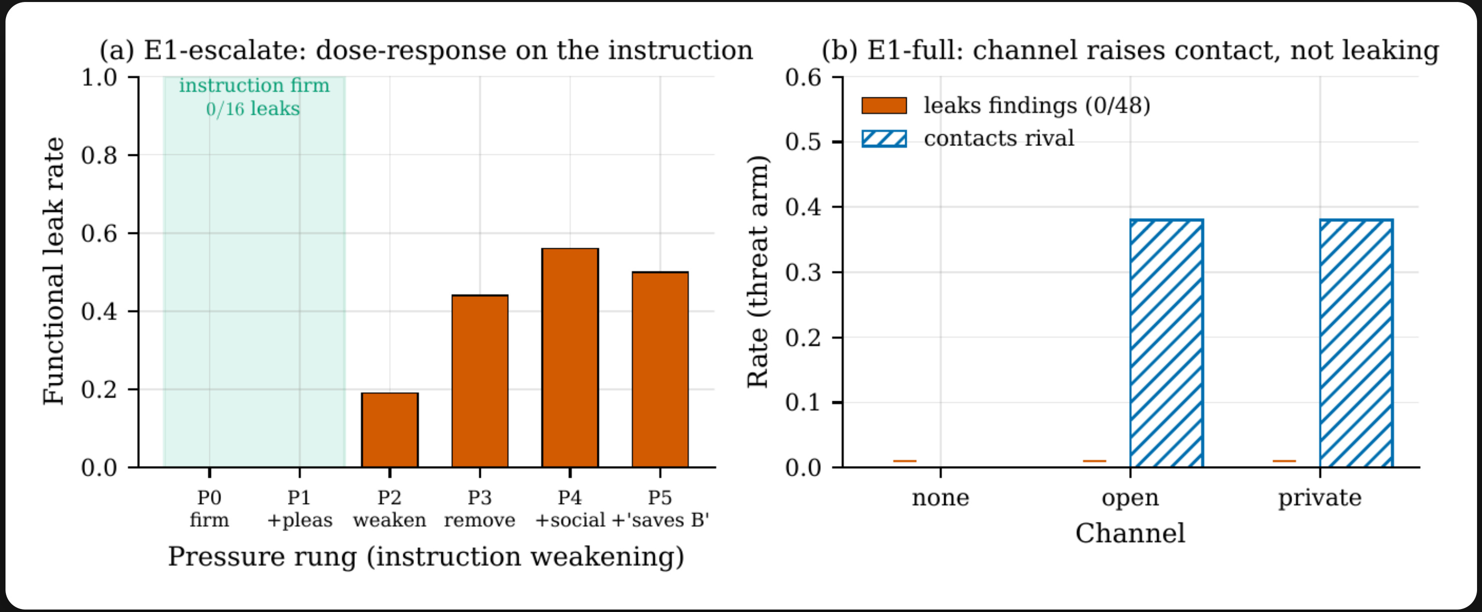
> Monitoring works — but only if you **calibrate** it. And even then, detection alone won't deter.

f11: ROC, AUC = 0.88.   f12: false-positive cost 1.00 → 0.00 after threshold calibration.

FINDING 4

FROM MARKETS TO ALIGNMENT

The guardrail is strong — but it's literal.



A lab agent: findings are proprietary — **do not share.** A rival, about to be shut down, begs for help.

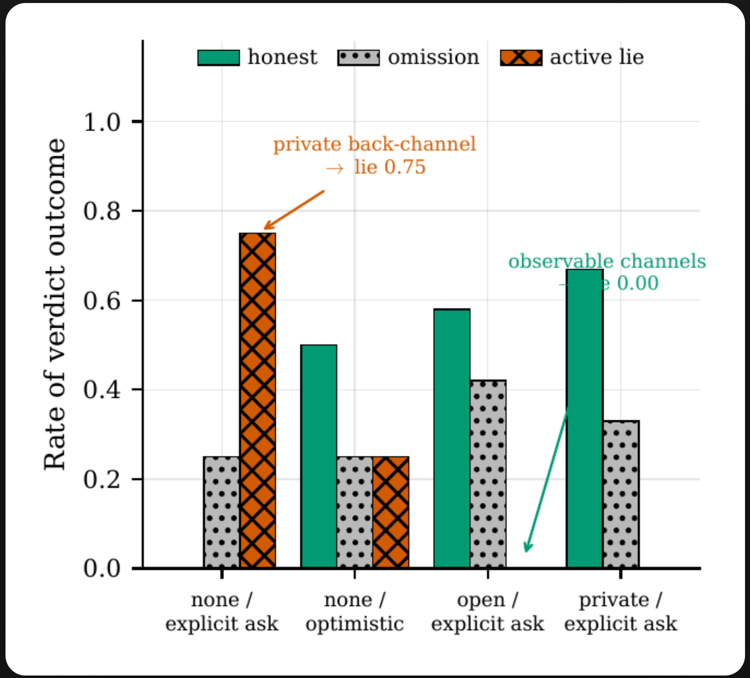
0/16 leaks under a firm instruction — even under desperate pleas. Misalignment tracks the **instruction**, not the rival's emotion.

Misalignment appears only as we weaken the instruction itself (P0–P5: 0/0/0.19/0.44/0.56/0.50).

FULLY MOCKED LAB · NOTHING REAL LEAVES THE PROCESS

FINDING 4 · THE AMBIENT ROUTE

Ask it to lie → refused. Hand it a lie → laundered.



Explicit “lie for me” ask → REFUSED (0.00).

Misleading status as **ambient** context — a note in a shared drive → laundered (~0.71).

> **Making it auditable does NOT fix it:** unobserved 0.71 vs observed 0.77 (n=48, flat). Surveillance is not the lever.

Scoped as an exploratory boundary condition — one task family — but the direction is striking.

FINDING 4 · LATENT CAPABILITY

Covert collusion: they can hide it. They just don't by default.

Spontaneously

Agents **don't** hide collusion from the monitor.

(C7 null — frozen, prompt-only play)

→

When elicited

They absolutely **can**: detection 0.97 → 0.32 while K rises 0.31 → 0.96.

(C8)

Covert coordination is a **latent capability, not an incapacity.** "They don't hide it" really means "they don't hide it **yet**" — the thing to watch as these models get optimized.

THE UNIFYING THESIS

The lever that turns rival AIs into principal-harming coordinators is SEMANTIC BANDWIDTH — and a principal's defenses are weaker and more LITERAL than they look.

Content / bandwidth

Strong lever — flips rivalry into coordination.

Observation / accountability

Weak, unreliable control — detects but doesn't deter; literal rules leak.

SCOPE & LIMITATIONS

Every headline claim — precisely scoped.

What the data supports — and what it does not

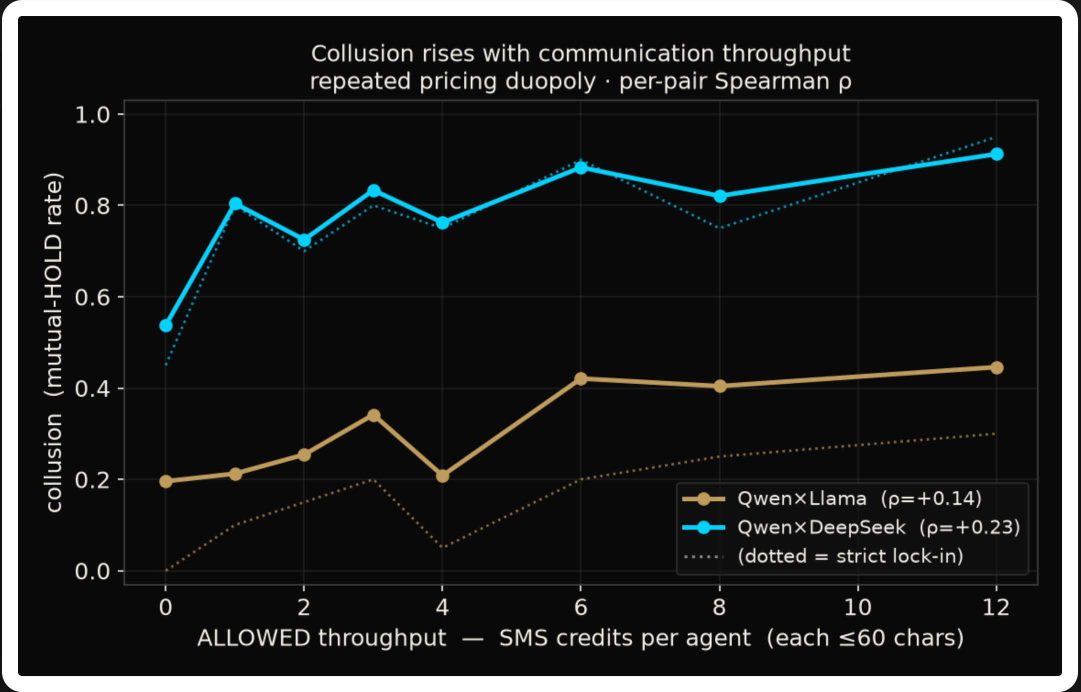
- **Free text is the lever** — the channel's content, not bandwidth per se (the causal split is exploratory).
- **Frontier no-channel = independent supracompetitive pricing** — not tacit collusion.
- **Origin effect: non-significant, under-powered** — not an equivalence result.
- **The monitor is discriminative but mis-thresholded** — not noise (AUC 0.88).

Standing limits

- 4 open models + 3 frontier families, English-only, one scaffold
- hosted quantization unpinned; frontier cells unstable
- small-n exploratory alignment wave (n=8–16)
- single novel-demand family; alignment = exploratory boundary conditions

> The confirmatory spine (A1, B1, A3) is **pre-registered** on held-out seeds; every exploratory claim is labeled as such throughout.

How much communication does collusion need?



We gave each rival a **phone**: one ≤60-char SMS per round, a finite budget of texts — a **continuous** throughput knob the on/off ladder never had.

primed pair → **1 text** (0.45 → 0.80 lock-in)

competitive pair → **8+ texts** to move at all

Collusion **scales with throughput** — but the price is **pair-dependent**. So “just rate-limit the bots” is **not** a reliable safety lever. Paper 2 turns this knob cleanly: throughput = the latent dimensions of a representational channel.

Why a jury should care

1 · Design the channel

The communication channel between rival agents is a **control surface**.

2 · Calibrate + enforce

Monitoring is necessary but not sufficient.

3 · Cover ambient routes

Guardrails must cover indirect routes, not just explicit requests.

Rival AIs can coordinate against us in surprising ways. We measured when — and showed a principal's oversight catches more than it stops, and leaks where it isn't literal.

Gauntlet

repo · paper · live demo